

DAVIDE PALLESCHI

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SUMMARY

Applied AI Researcher specializing in the **evaluation and application** of Large Language Models (LLMs). My work centers on **designing and conducting experiments** to improve AI product capabilities, including **comparative analysis and parameter-efficient fine-tuning (LoRA/PEFT)** of open-source models. I am skilled in **in-depth LLM benchmarking and evaluation** (tool use, reasoning, accuracy) and prototyping advanced agentic systems (**LangGraph**, RAG) to solve complex tasks. I thrive in R&D teams collaborating to **transform research findings into practical, cutting-edge solutions**.

EXPERIENCE

DeepPlants

Rome, Italy

AI Engineer

09/2024 – Present

- Prototyping and researching advanced **multi-agent architectures (LangGraph)** for complex automation and multi-step user objectives.
- **Conducted comparative analysis and fine-tuning (LoRA/PEFT)** on open-source LLMs to identify optimal models for product-specific reasoning tasks.
- **Designed and executed experiments** to benchmark LLM interaction patterns, focusing on tool use, stepwise reasoning, and output reliability.
- Engineered context management systems (retrieval augmentation, memory persistence, grounding) for high-reliability agent orchestration.

Sapienza S.r.l.

Rome, Italy

Software Developer / Web Designer

04/2018 – 07/2024

- Developed automation scripts in Python and supported internal IT systems, enabling efficiency for daily operations.

EDUCATION

Sapienza University of Rome — *M.Sc., AI and Robotics*

09/2020 – 03/2024

- Grade: 102/110 • Erasmus: Polytechnic University of Valencia (2022)
- Thesis: *Privacy-Oriented Object Detection (low-resolution video, YOLOv8)*

University of Rome Tor Vergata — *B.Sc., Computer Engineering*

09/2015 – 08/2020

- Grade: 101/110 • Erasmus: Imperial College London (2018–2019)

SKILLS

LLM Fine-Tuning & NLP: Fine-tuning (LoRA/PEFT), Hugging Face **Transformers**, text classification, traditional NLP

LLM Research & Evaluation: **Designing experiments**, benchmarking agent/LLM stacks, output reliability analysis, API integration, multi-GPU experiments

Agent Systems & Context: Multi-agent orchestration (LangGraph), memory management, retrieval augmentation (RAG), prompt engineering

Stack Prototyping: **Python, PyTorch**, FastAPI, Docker, vLLM, Streamlit

Languages: Italian (Native), English (Professional)